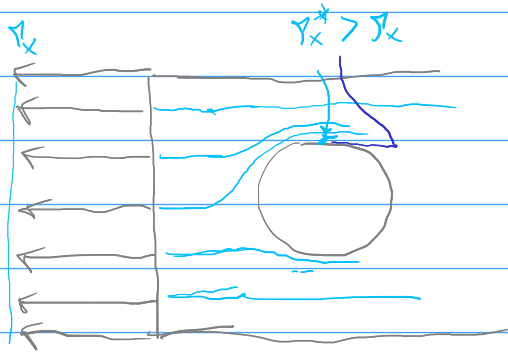
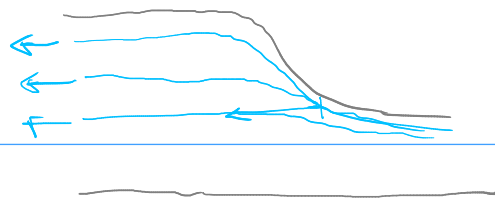
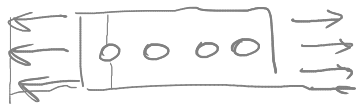
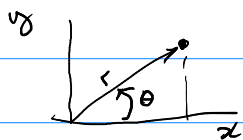


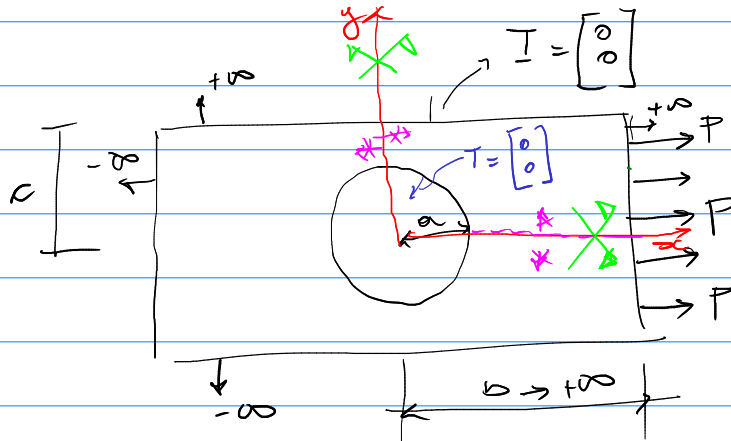


$r > 0$

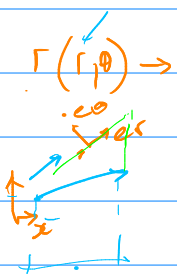
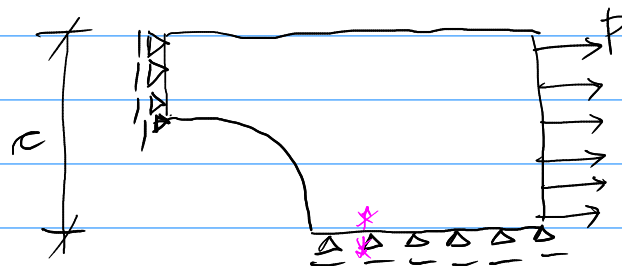


$$\begin{cases} x = r \cdot \cos \theta \\ y = r \cdot \sin \theta \\ z = z \end{cases}$$

$$\begin{cases} \theta = \arctan\left(\frac{y}{x}\right) \\ r^2 = x^2 + y^2 \\ z = z \end{cases}$$

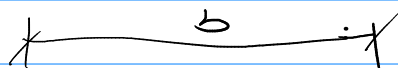


$$\underline{I} = \begin{bmatrix} P & 0 \\ 0 & 0 \end{bmatrix}$$



1)

$$P \begin{bmatrix} \gamma_r(r, \theta) & \gamma_\theta(r, \theta) \\ \gamma_\theta(r, \theta) & \gamma_r(r, \theta) \end{bmatrix} P$$



$$\begin{bmatrix} \gamma_x(x, y) & \gamma_{xy} \\ \gamma_{xy} & \gamma_y \end{bmatrix}$$

$$\gamma_x(x, y)$$

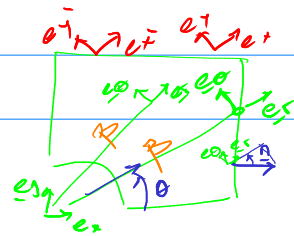
$$\underline{n} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\underline{I} = \underline{\gamma} \cdot \underline{n}$$

2)

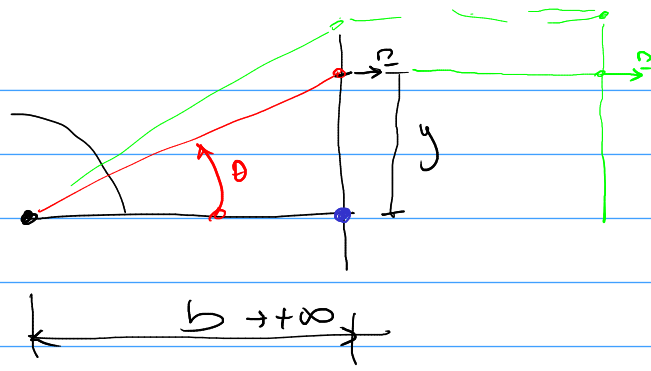
$$\begin{bmatrix} \gamma_r(r, \theta) & \gamma_\theta(r, \theta) \end{bmatrix}$$

$$\lim_{\Delta \rightarrow +\infty} \underline{I} = \underline{\gamma} \cdot \underline{n}$$

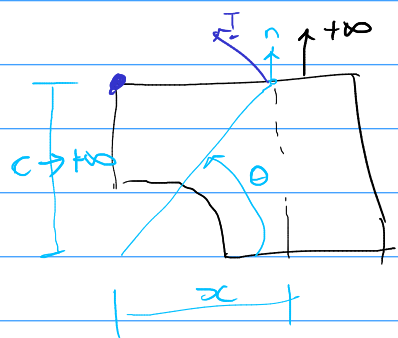


$$\underline{n} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

b)

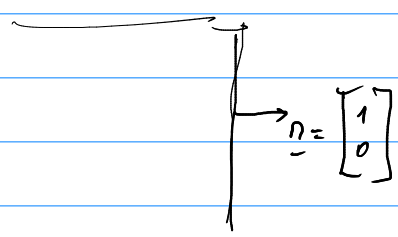


$$\theta = \lim_{b \rightarrow +\infty} \arctan\left(\frac{y}{b}\right) = 0$$



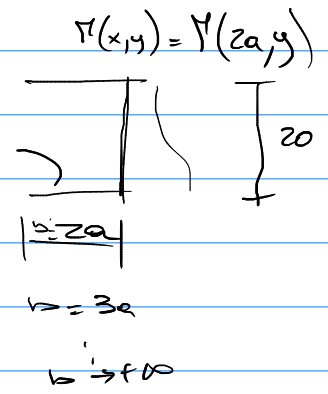
$$\theta = \arctan\left(\frac{c}{x}\right) = \frac{\pi}{2}$$

$\rightarrow x$ es de en cada punto

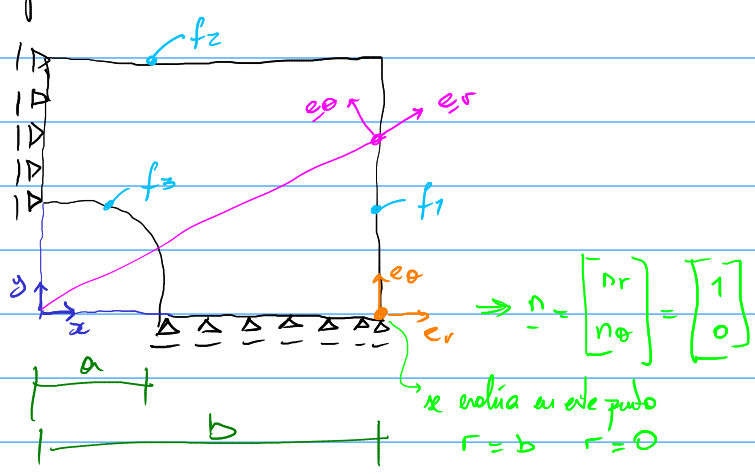


$$P = \begin{bmatrix} P_x & \partial_{xy} \\ \partial_{xy} & P_y \end{bmatrix}$$

$$T = P \cdot n = \begin{bmatrix} P_x \\ \partial_{xy} \end{bmatrix}$$



a) Verificación de condiciones de borde



f1)

$$T(r, \theta) \rightarrow T(b, 0) = T(b, 0) \cdot n = \begin{bmatrix} P_r & \partial_{r\theta} \\ \partial_{r\theta} & P_\theta \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} P_r(b, 0) \\ \partial_{r\theta}(b, 0) \end{bmatrix}$$

$$\sigma_r = \frac{p}{2} \left(1 - \frac{a^2}{r^2} \right) \left[1 + \left(1 - 3 \frac{a^2}{r^2} \right) \cos 2\theta \right]$$

$$\lim_{b \rightarrow +\infty} \sigma_r(b, 0) = \frac{p}{2} \left(1 - \frac{a^2}{b^2} \right) \cdot \left[1 + \left(1 - 3 \frac{a^2}{b^2} \right) \cos(2 \cdot 0) \right] = \frac{p}{2} \cdot 2 = p$$

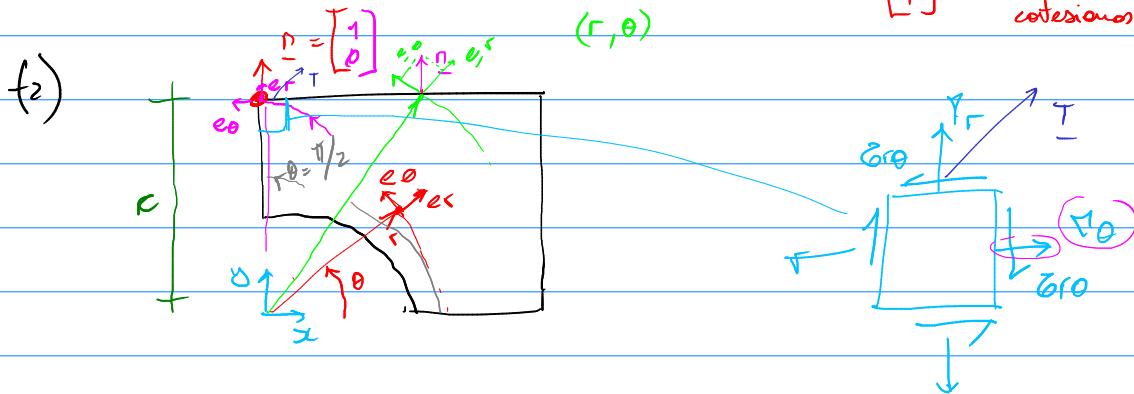
$$\tau_{r\theta} = -\frac{p}{2} \left(1 - \frac{a^2}{r^2} \right) \left(1 + 3 \frac{a^2}{r^2} \right) \sin 2\theta$$

$$\lim_{b \rightarrow +\infty} \tau_{r\theta}(b, 0) = -\frac{p}{2} \left(1 - \frac{a^2}{b^2} \right) \cdot \left(1 + 3 \frac{a^2}{b^2} \right) \cdot \sin(2 \cdot 0) = 0$$

$$T = \begin{bmatrix} p \\ 0 \end{bmatrix} \Rightarrow \text{Vérifier le CB}$$

$$\lim_{b \rightarrow +\infty} \sigma_\theta(b, 0) = \frac{p}{2} \cdot \left[1 - \frac{a^2}{b^2} - \left(1 + 3 \frac{a^2}{b^2} \right) \cdot \cos(2 \cdot 0) \right] = 0$$

$$n = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \text{condiciones de contorno en } x \text{ e } y$$



$$n = \begin{bmatrix} n_r \\ n_\theta \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$T = \begin{bmatrix} \sigma_{rr} & \sigma_{r\theta} \\ \sigma_{r\theta} & \sigma_{\theta\theta} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \sigma_{rr} \\ \sigma_{r\theta} \end{bmatrix}$$

$$\theta = \pi/2 \quad r = c$$

$$\lim_{c \rightarrow +\infty} \sigma_\theta(c, \pi/2) = \dots$$

$$= 0$$

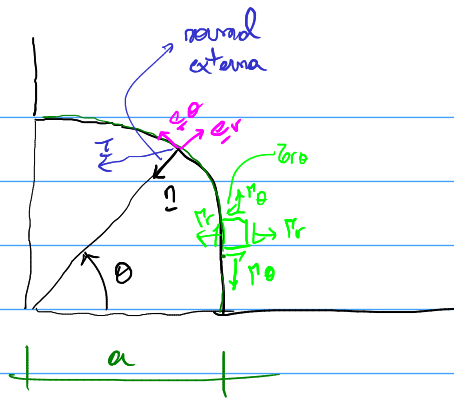
$$\lim_{c \rightarrow +\infty} \tau_{r\theta}(c, \pi/2) = \dots$$

$$= 0$$

$\Rightarrow$ Cumple con los CB

— • —

$$\lim_{c \rightarrow +\infty} \sigma_\theta(c, \pi/2) = \frac{p}{2} \left[1 + \frac{a^2}{c^2} - \left(1 + 3 \frac{a^2}{c^2} \right) \cdot \cos \left(2 \cdot \frac{\pi}{2} \right) \right] = p$$



$$\underline{n} = \begin{bmatrix} n_r \\ n_\theta \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$T = \begin{bmatrix} T_r \\ T_\theta \end{bmatrix} = \begin{bmatrix} \sigma_{rr} & \sigma_{r\theta} \\ \sigma_{\theta r} & \sigma_{\theta\theta} \end{bmatrix} \cdot \begin{pmatrix} -1 \\ 0 \end{pmatrix} = - \begin{bmatrix} \sigma_{rr} \\ \sigma_{r\theta} \end{bmatrix}$$

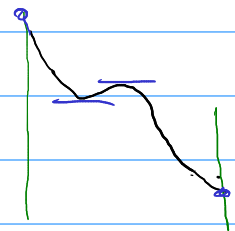
$$r = a \quad \theta \in [0, \pi/2)$$

$$\left. \begin{aligned} \sigma_{rr}(a, \theta) &= \frac{P}{2} \left(1 - \frac{a^2}{a^2} \right) \left[1 + \left(1 - 3 \frac{a^2}{a^2} \right) \cos(2\theta) \right] = 0 \\ \sigma_{r\theta}(a, \theta) &= -\frac{P}{2} \left(1 - \frac{a^2}{a^2} \right) \left(1 + 3 \frac{a^2}{a^2} \right) \sin(2\theta) = 0 \end{aligned} \right\} \Rightarrow T = 0 \Rightarrow \text{Verif. a} \\ \text{los CB}$$

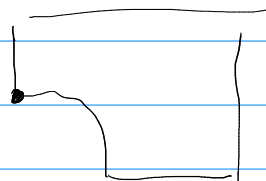
— o —

$$\begin{aligned} \sigma_{\theta\theta}(a, \theta) &= \frac{P}{2} \left[1 + \frac{a^2}{a^2} - \left(1 + 3 \frac{a^4}{a^4} \right) \cos(2\theta) \right] = \frac{P}{2} (2 - 4 \cos(2\theta)) = \\ &= P (1 - 2 \cos(2\theta)) \end{aligned}$$

1D)



b)



$$\frac{\partial \sigma_{\theta\theta}}{\partial r} = 0 \Rightarrow r^2 = 6a^2 \cos(2\theta)$$

$$\frac{\partial \sigma_{\theta\theta}}{\partial \theta} = 0 \Rightarrow \sin(2\theta) = 0 \rightarrow \begin{aligned} &\text{dentro de } \theta \in [0, \pi/2) \\ &\rightarrow \theta = 0 \\ &\rightarrow \theta = \pi/2 \end{aligned}$$

$$r^2 = 6a^2 \cdot \cos(2\theta) \Rightarrow r = \sqrt{6}a \cdot \sqrt{\cos(2\theta)}$$

$\rightarrow +1 \Rightarrow \theta = 0$
 $r = \sqrt{6} \cdot a$

$$V_0(\sqrt{6}a, 0) = P/24$$

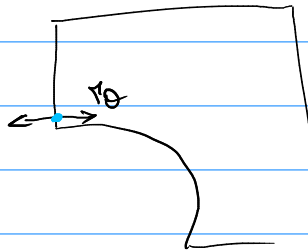
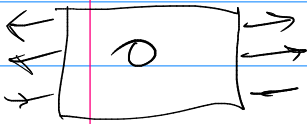
En la frontera $f_1 \rightarrow V_0 = 0$

" $f_2 \rightarrow V_0 = P$

En la frontera $f_3 \rightarrow V_0(a, \theta) = P(1 - 2\cos(2\theta))$

$$\frac{\partial V_0(a, \theta)}{\partial \theta} = 0 \Rightarrow \sin(2\theta) = 0 \begin{cases} \theta = 0 \\ \theta = \pi/2 \end{cases}$$

$$V_0(a, \theta) = \begin{cases} \theta = 0 & -P \\ \theta = \pi/2 & 3P \end{cases}$$



$$V_{0\max} = 3P$$

$(\theta = \pi/2 \text{ y } r = a)$

Para c y d determinar la ubicación del máximo gráficamente y luego analíticamente los valores máximos.

$$\begin{cases} \frac{\partial V_{0\max}}{\partial \theta} = 0 \\ \frac{\partial V_{0\max}}{\partial r} = 0 \end{cases}$$